



Artificial Intelligence (AI-2002)

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ML- Classification and Regression

Gentle Reminder

***“Switch Off” your Mobile Phone Or Switch
Mobile Phone to “Silent Mode”***



Contents



1. Classification Vs. Prediction

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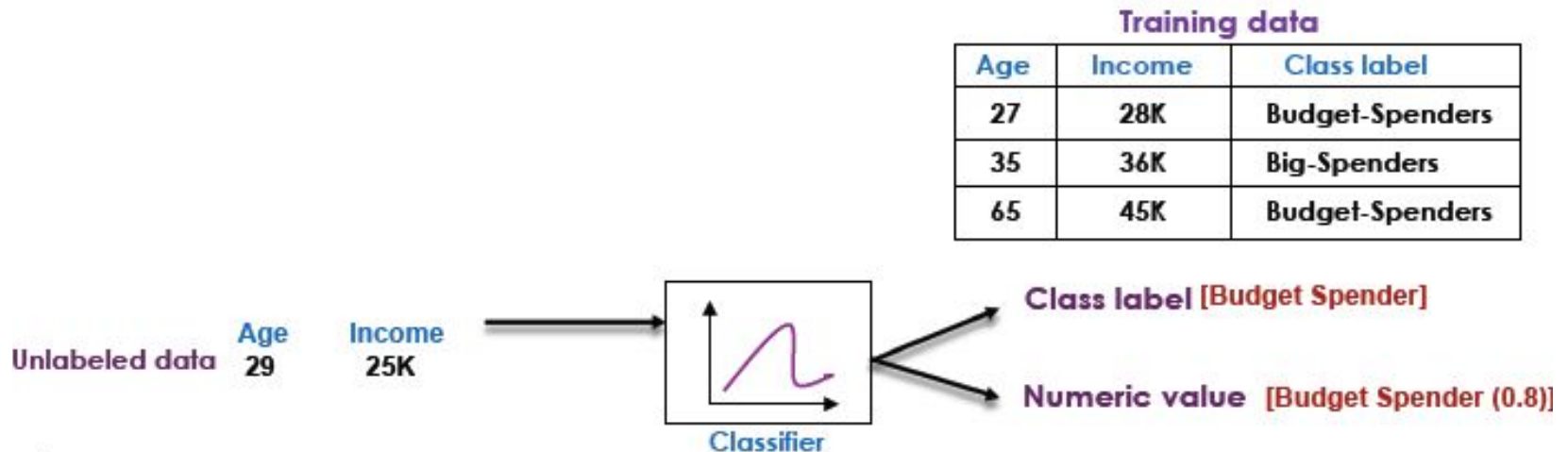
7. Classifier Accuracy

Definition

❖ Classification is also called **Supervised Learning**

❖ **Supervision**

- The training data (observations, measurements, etc) are used to learn a classifier
- The training data are **labeled data**
- New data (**unlabeled**) are **classified** Using the training data



❖ **Principle**

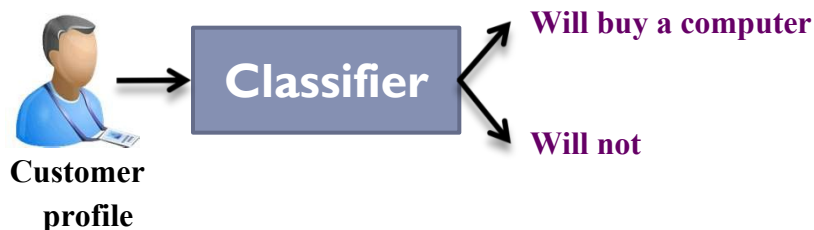
- Construct models (functions) based on some training examples
- Describe and distinguish classes or concepts for future prediction
- Predict some unknown class labels

Classification vs. Prediction

❖ Classification

- Predicts categorical class labels (discrete or nominal)
- Use labels of the training data to classify new data

❖ Example



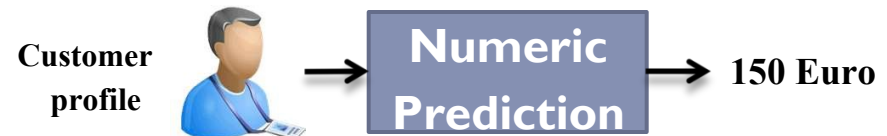
- A model or classifier is constructed to predict **categorical labels** such as “safe” or “risky” for a loan application data.

❖ Prediction

- ❖ Models continuous-valued functions, i.e., predicts unknown or missing values

❖ Example

- ❖ A marketing manager would like to predict how much a given customer will spend during a sale



- ❖ Unlike classification, it provides ordered values
- ❖ **Regression analysis is used for prediction**
- ❖ Prediction is a short name for
- ❖ **numeric prediction**

Applications

❖ Typical applications

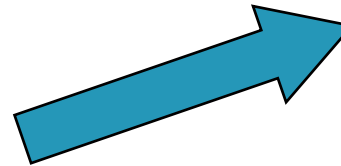
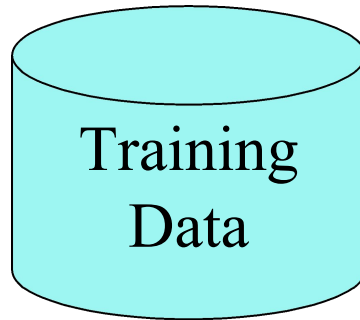
- Credit approval
- Target marketing
- Medical diagnosis
- Treatment effectiveness analysis
- Fraud detection
- Defective parts identification in manufacturing
- Crime zoning



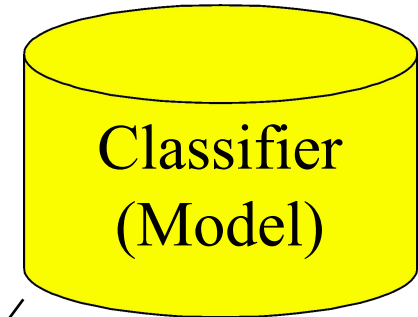
Classification—A Three-Step Process

- ❖ **Model construction:** describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- ❖ **Test Classifier (Estimate accuracy of the model)**
 - The known label of **test sample** is compared with the classified result from the model
 - Accuracy rate is the **percentage** of test set samples that are correctly classified by the model
 - Test set is **independent** of training set, otherwise **over-fitting** will occur
- ❖ **Model usage:** for classifying future or unknown objects
 - If the accuracy is acceptable, use the model to **classify data** tuples whose class labels are not known

Process (1): Model Construction



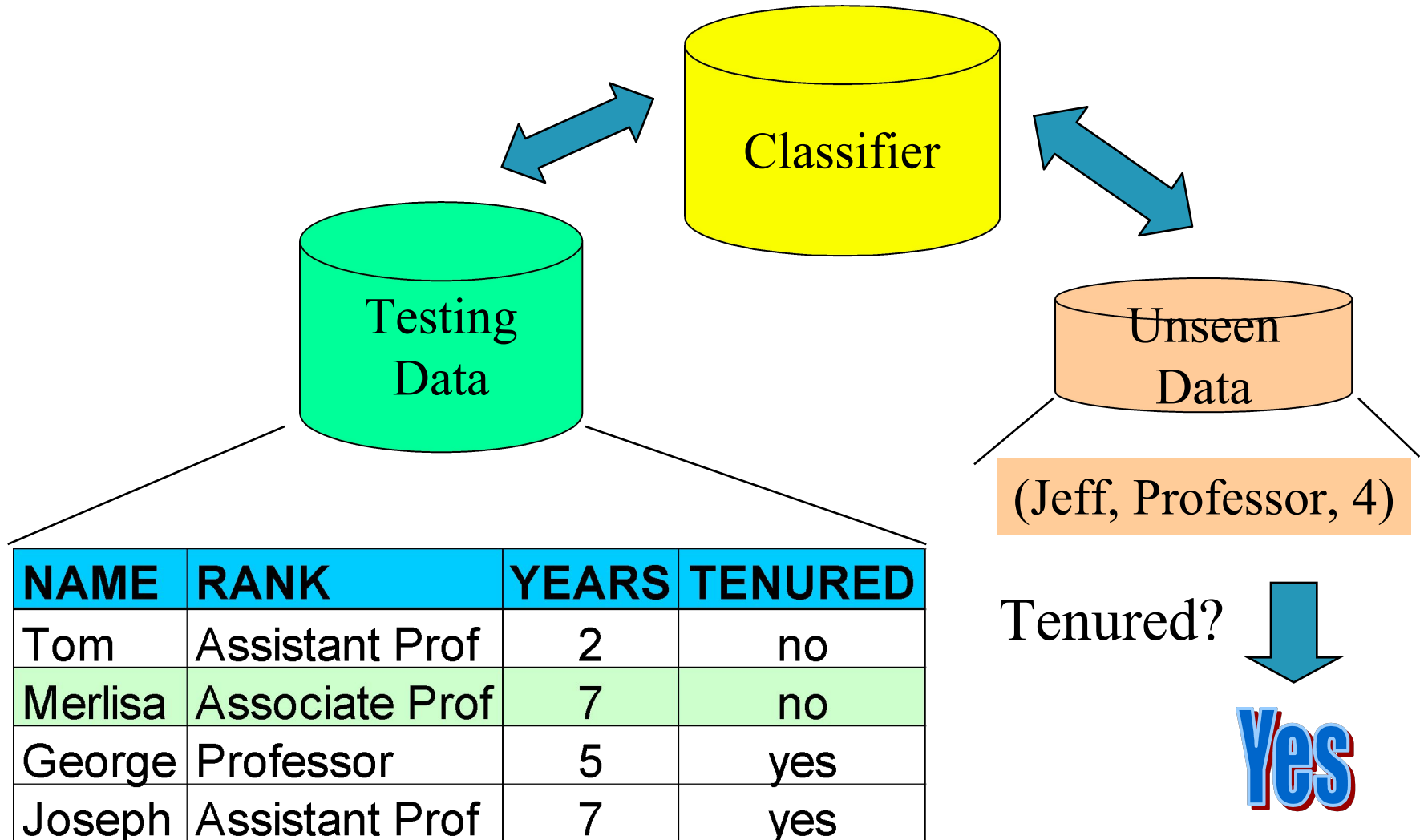
Classification
Algorithms



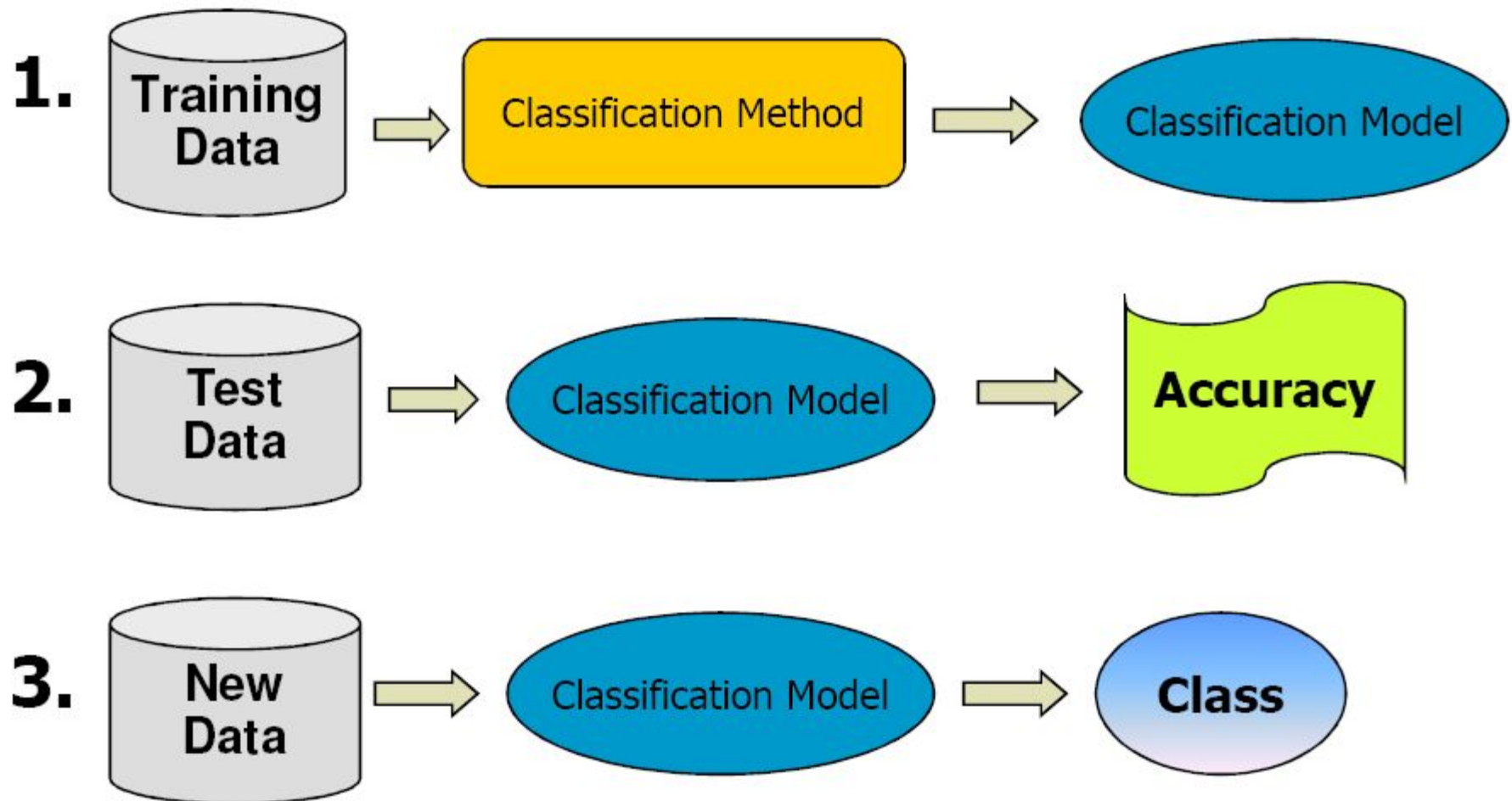
NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

IF rank = 'professor'
OR years > 6
THEN tenured = 'yes'

Process (2&3): Test and Using the Model in Prediction



Classification—A Three-Step Process



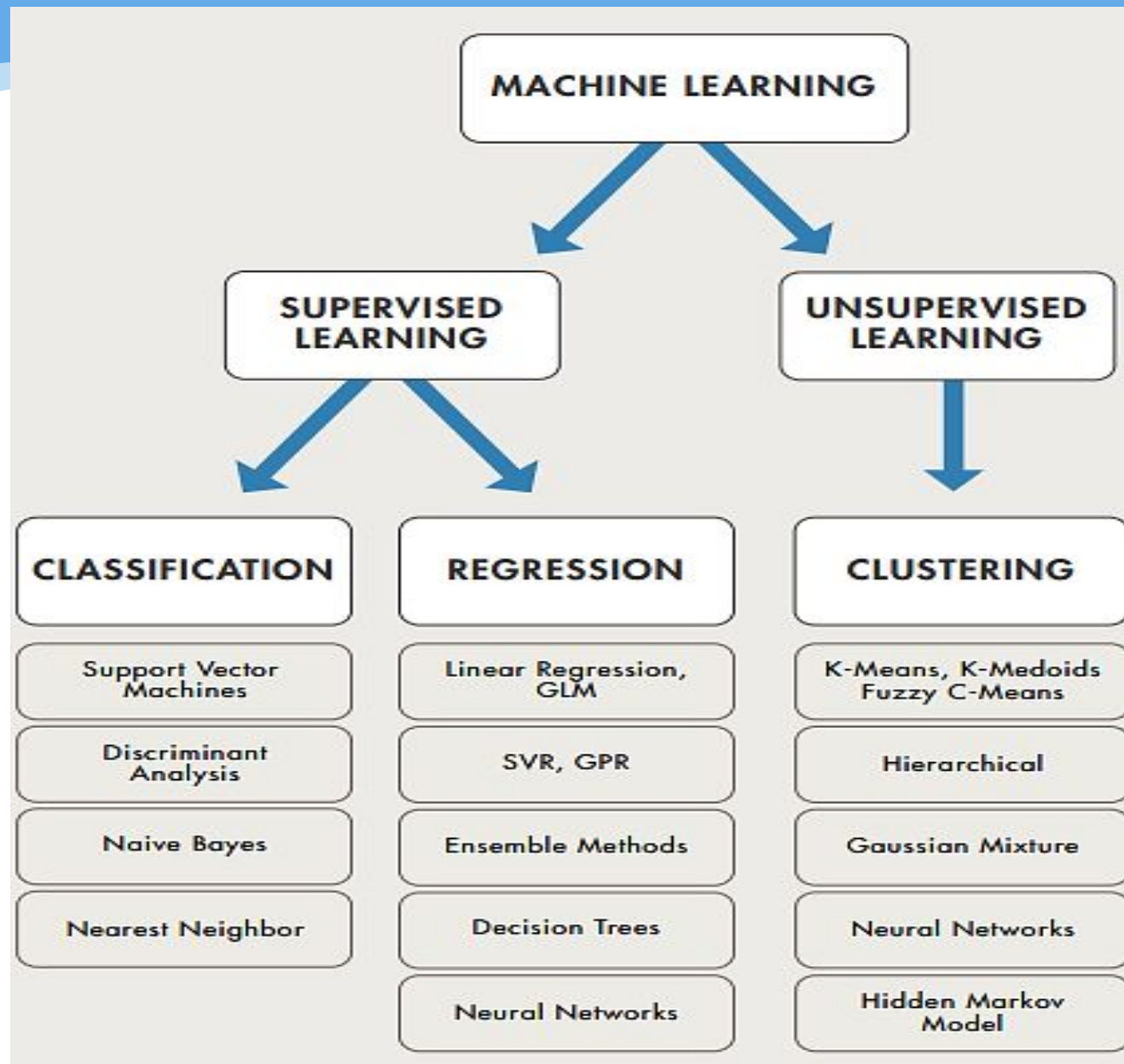
Supervised vs. Unsupervised Learning

❖ Supervised learning (classification)

- Supervision: The training data (observations, measurements, etc.) are accompanied by labels indicating the class of the observations
- New data is classified based on the training set

❖ Unsupervised learning (clustering)

- The class labels of training data is unknown
- Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data



Issues: Data Preparation

- ❖ Data cleaning
 - Preprocess data in order to reduce noise and handle missing values
- ❖ Relevance analysis (feature selection)
 - Remove the irrelevant or redundant attributes
- ❖ Data transformation
 - Generalize and/or normalize data

Issues: Evaluating Classification Methods

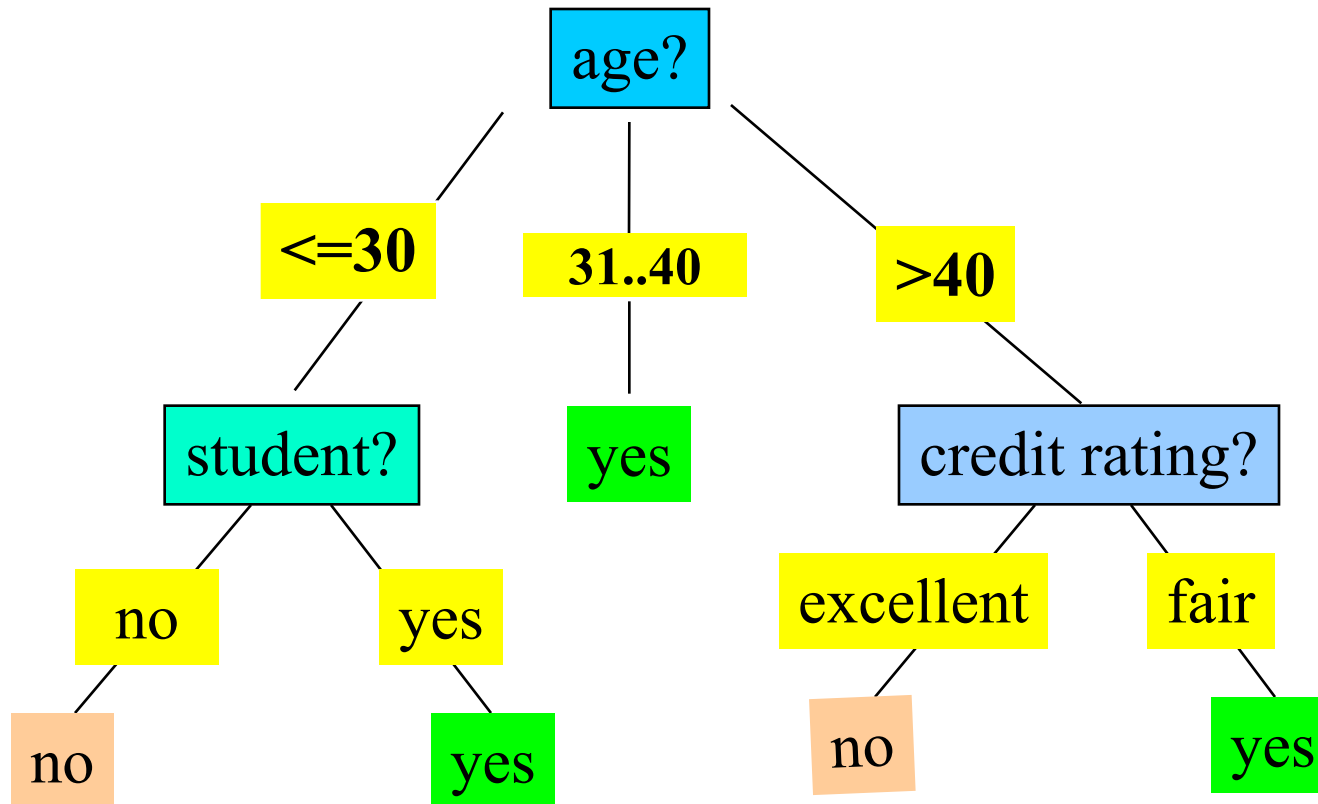
- ❖ Accuracy
 - classifier accuracy: predicting class label
 - predictor accuracy: guessing value of predicted attributes
- ❖ Speed
 - time to construct the model (training time)
 - time to use the model (classification/prediction time)
- ❖ Robustness: handling noise and missing values
- ❖ Scalability: efficiency in disk-resident databases
- ❖ Interpretability
 - understanding and insight provided by the model
- ❖ Other measures, e.g., goodness of rules, such as decision tree size or compactness of classification rules

Decision Tree Induction: Training Dataset

This
follows an
example
of (Buy
Computer)

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Output: A Decision Tree for “*buys_computer*”



Algorithm for Decision Tree Induction

Basic algorithm (a greedy algorithm)

1. Tree is constructed in a **top-down recursive divide-and-conquer manner**
2. At start, all the training examples are at the root
3. **Attributes are categorical** (if continuous-valued, they are **discretized in advance**)
4. Examples are partitioned recursively based on selected attributes
5. Test attributes are selected **on the basis of a heuristic or statistical measure** (e.g., **information gain, gain ratio, gini index**)

❖ Conditions for **stopping partitioning**

- All samples for a given node belong to the same class
- There are no remaining attributes for further partitioning
 - **majority voting** is employed for classifying the leaf
- There are no samples left

Attribute Selection Measure: Information Gain (ID3/C4.5)

- Select the attribute with the **highest information gain**
- Let p_i be the probability that an arbitrary tuple in D belongs to class C_i , estimated by $|C_{i,D}|/|D|$
- **Expected information** (entropy) needed to classify a tuple in D :

$$Info(D) = -\sum_{i=1}^m p_i \log_2(p_i)$$

- **Information** needed (after using A to split D into v partitions) to classify D :

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times I(D_j)$$

- **Information gained** by branching on attribute A

$$Gain(A) = Info(D) - Info_A(D)$$

Attribute Selection: Information Gain

g Class P: buys_computer = "yes"

g Class N: buys_computer = "no"

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

age	p_i	n_i	$I(p_i, n_i)$
≤ 30	2	3	0.971
31...40	4	0	0
> 40	3	2	0.971

$$Info_{age}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) = 0.694$$

$\frac{5}{14} I(2,3)$ means "age ≤ 30 " has 5 out of 14 samples, with 2 yes'es and 3 no's. Hence

$$Gain(age) = Info(D) - Info_{age}(D) = 0.246$$

Similarly,

$$Gain(income) = 0.029$$

$$Gain(student) = 0.151$$

$$Gain(credit_rating) = 0.048$$

age	income	student	credit rating	buys computer
≤ 30	high	no	fair	no
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31...40	high	no	fair	yes
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Attribute Selection: Information Gain

RID	age	income	student	Credit_rating	Class: buys_computer
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$$C_1 = \text{Yes} = 9$$

$$C_2 = \text{No} = 5$$

1- compute $info(D)$

$$= - \sum_{i=1}^m p_i \log_2(p_i)$$

$$= - \frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right)$$

$$= 0.940 \text{ bits}$$

2- compute $info_{age}(D)$

$$= \sum_{j=1}^v \frac{|D_j|}{|D|} \times info(D_j)$$

$$= \frac{5}{14} \times \left(-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right)$$

$$+ \frac{4}{14} \times \left(-\frac{4}{4} \log_2 \frac{4}{4} \right)$$

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$$+ \frac{5}{14} \times \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right)$$

$$= 0.694 \text{ bits}$$

3- compute $Gain(age) = 0.94 - 0.694 = 0.246 \text{ bits}$

Attribute Selection: Information Gain

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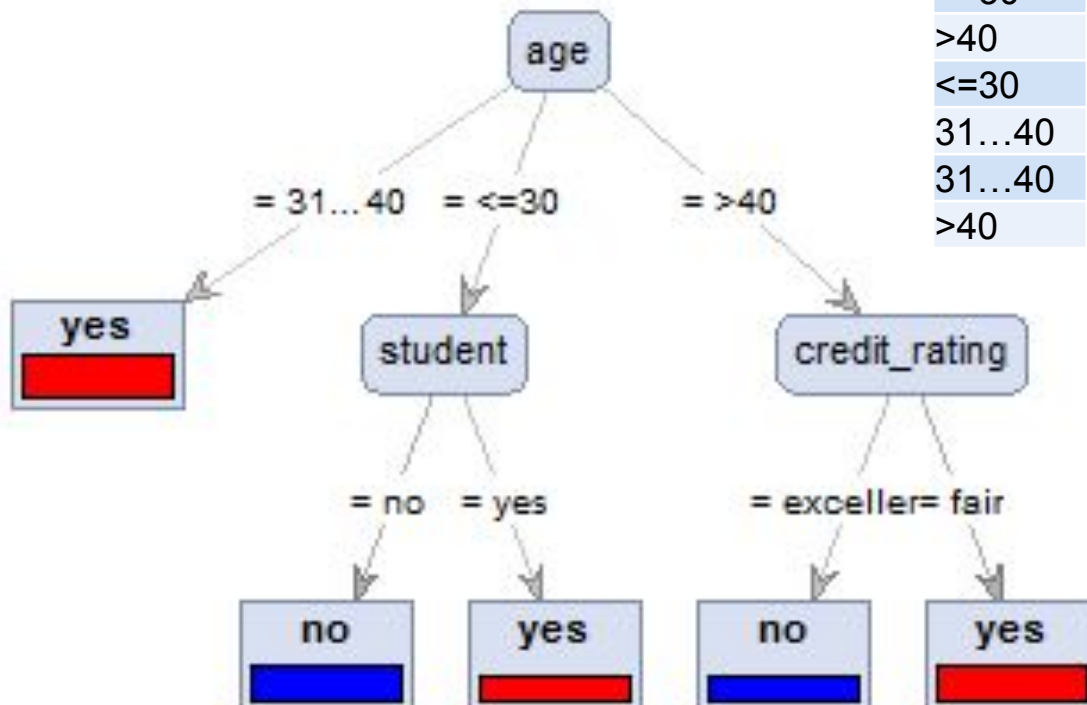
Gain(age) = 0.246 bits
Gain(income) = 0.029 bits
Gain(student) = 0.151 bits
Gain(credit_rating) = 0.048 bits

Gain(age) has highest information gain

Attribute Selection: Information Gain

❖ Training data set:
Buys_computer

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
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<=30	medium	no	fair	no
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>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no



Practice : Training Data

No	OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	PLAY
1	Sunny	Hot	High	FALSE	No
2	Sunny	Hot	High	TRUE	No
3	Cloudy	Hot	High	FALSE	Yes
4	Rainy	Mild	High	FALSE	Yes
5	Rainy	Cool	Normal	FALSE	Yes
6	Rainy	Cool	Normal	TRUE	Yes
7	Cloudy	Cool	Normal	TRUE	Yes
8	Sunny	Mild	High	FALSE	No
9	Sunny	Cool	Normal	FALSE	Yes
10	Rainy	Mild	Normal	FALSE	Yes
11	Sunny	Mild	Normal	TRUE	Yes
12	Cloudy	Mild	High	TRUE	Yes
13	Cloudy	Hot	Normal	FALSE	Yes
14	Rainy	Mild	High	TRUE	No

Practice: Training Data (Entropy Calculations)

❖ Entropy **Total**

$$Entropy(Total) = (-\frac{4}{14} * \log_2(\frac{4}{14})) + (-\frac{10}{14} * \log_2(\frac{10}{14}))$$

$$Entropy(Total) = 0.863120569$$

❖ Entropy (**Outlook**)

$$Entropy(Cloudy) = (-\frac{0}{4} * \log_2(\frac{0}{4})) + (-\frac{4}{4} * \log_2(\frac{4}{4})) = 0.000000000$$

$$Entropy(Rainy) = (-\frac{1}{5} * \log_2(\frac{1}{5})) + (-\frac{4}{5} * \log_2(\frac{4}{5})) = 0.721928095$$

$$Entropy(Sunny) = (-\frac{3}{5} * \log_2(\frac{3}{5})) + (-\frac{2}{5} * \log_2(\frac{2}{5})) = 0.970950594$$

❖ Entropy (**Temperature**)

$$Entropy(Cool) = (-\frac{0}{4} * \log_2(\frac{0}{4})) + (-\frac{4}{4} * \log_2(\frac{4}{4})) = 0.000000000$$

$$Entropy(Hot) = (-\frac{2}{4} * \log_2(\frac{2}{4})) + (-\frac{2}{4} * \log_2(\frac{2}{4})) = 1.000000000$$

$$Entropy(Mild) = (-\frac{2}{6} * \log_2(\frac{2}{6})) + (-\frac{4}{6} * \log_2(\frac{4}{6})) = 0.918295834$$

❖ Entropy (**Humidity**)

$$Entropy(High) = (-\frac{4}{7} * \log_2(\frac{4}{7})) + (-\frac{3}{7} * \log_2(\frac{3}{7})) = 0.985228136$$

$$Entropy(Normal) = (-\frac{0}{7} * \log_2(\frac{0}{7})) + (-\frac{7}{7} * \log_2(\frac{7}{7})) = 0.000000000$$

❖ Entropy (**Windy**)

$$Entropy(False) = (-\frac{2}{8} * \log_2(\frac{2}{8})) + (-\frac{6}{8} * \log_2(\frac{6}{8})) = 0.811278124$$

$$Entropy(True) = (-\frac{4}{6} * \log_2(\frac{4}{6})) + (-\frac{2}{6} * \log_2(\frac{2}{6})) = 0.918295834$$

Bayesian Classification: Why?

- ❖ A **statistical classifier**: performs probabilistic prediction, i.e., predicts class membership probabilities
- ❖ **Foundation**: Based on Bayes' Theorem.
- ❖ **Performance**: A simple Bayesian classifier, naïve Bayesian classifier, has comparable performance with decision tree and selected neural network classifiers
- ❖ **Incremental**: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data
- ❖ **Standard**: Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayesian Theorem: Basics

- ❖ Let $\mathbf{X}$ be a data sample (“*evidence*”): class label is unknown
- ❖ Let H be a *hypothesis* that X belongs to class C
- ❖ Classification is to determine $P(H|\mathbf{X})$, the probability that the hypothesis holds given the observed data sample $\mathbf{X}$
- ❖ $P(H)$ (*prior probability*), the initial probability
 - E.g., $\mathbf{X}$ will buy computer, regardless of age, income, ...
- ❖ $P(\mathbf{X})$: probability that sample data is observed
- ❖ $P(\mathbf{X}|H)$ (*posteriori probability*), the probability of observing the sample $\mathbf{X}$, given that the hypothesis holds
 - E.g., Given that $\mathbf{X}$ will buy computer, the prob. that X is 31..40, medium income

Bayesian Theorem

- ❖ Given training data $\mathbf{X}$, *posteriori probability of a hypothesis* H , $P(H|\mathbf{X})$, follows the Bayes theorem

$$P(H | \mathbf{X}) = \frac{P(\mathbf{X} | H)P(H)}{P(\mathbf{X})}$$

- ❖ Informally, this can be written as
posteriori = likelihood x prior/evidence
- ❖ Predicts $\mathbf{X}$ belongs to C_2 iff the probability $P(C_i|\mathbf{X})$ is the highest among all the $P(C_k|\mathbf{X})$ for all the k classes
- ❖ Practical difficulty: require initial knowledge of many probabilities, significant computational cost

Towards Naïve Bayesian Classifier

- ❖ Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n -D attribute vector $\mathbf{X} = (x_1, x_2, \dots, x_n)$
- ❖ Suppose there are m classes $C_1, C_2, \dots, C_m$.
- ❖ Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$
- ❖ This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- ❖ Since $P(\mathbf{X})$ is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$

needs to be maximized

Derivation of Naïve Bayes Classifier

- ❖ A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(\mathbf{X} | C_i) = \prod_{k=1}^n P(x_k | C_i) = P(x_1 | C_i) \times P(x_2 | C_i) \times \dots \times P(x_n | C_i)$$

- ❖ This greatly reduces the computation cost: Only counts the class distribution
- ❖ If A_k is categorical, $P(x_k | C_i)$ is the # of tuples in C_i having value x_k for A_k divided by $|C_{i,D}|$ (# of tuples of C_i in D)
- ❖ If A_k is continuous-valued, $P(x_k | C_i)$ is usually computed based on Gaussian distribution with a mean μ and standard deviation σ

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

and $P(x_k | C_i)$ is

$$P(\mathbf{X} | C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i})$$

Naïve Bayesian Classifier: Training Dataset

Class:

C_1 :buys_computer = 'yes'

C_2 :buys_computer = 'no'

Data to be classified:

X = (age ≤ 30,
income = medium,
student = yes,
credit_rating = fair)

X ? buy computer?

age	income	student	credit_rating	com
≤30	high	no	fair	no
≤30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayesian Classifier: An Example

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

❖ $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

❖ Likelihood of “yes” = $\frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14}$
= **0.02821**

❖ Likelihood of “no” = $\frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14}$
= **0.0068**

❖ Turn into probability by normalizing likelihood values
probability of “yes” = $0.02821 / (0.02821 + 0.0068)$
= **0.8057**

❖ probability of “no” = $0.0068 / (0.02821 + 0.0068)$
= **0.1942**

age			Income			Student			Credit Rating			Buys_Computer	
	Yes	No		Yes	No		Yes	No		Yes	No	Yes	No
<=30	2/9	3/5	High	2/9	2/5	Yes	6/9	1/5	Excellent	3/9	3/5	9/14	5/14
31..40	4/9	0/5	Medium	4/9	2/5	No	3/9	4/5	Fair	6/9	2/5		
>40	3/9	2/5	Low	3/9	1/5								

probability of “yes” is 0.8057 therefore X belongs to (“buy_computer = yes”)

Naïve Bayesian Classifier: Exercise

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

The given data belongs to which class (mammals/non-mammals)?

Use Naïve Bayesian classifier.

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

Naïve Bayes Classifier: Comments

❖ Advantages

- Easy to implement
- Good results obtained in most of the cases

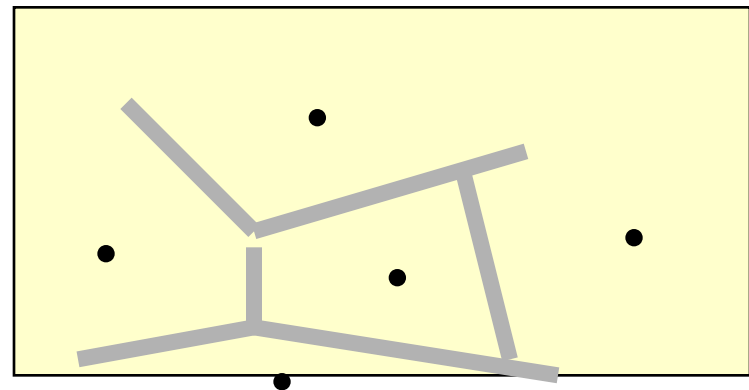
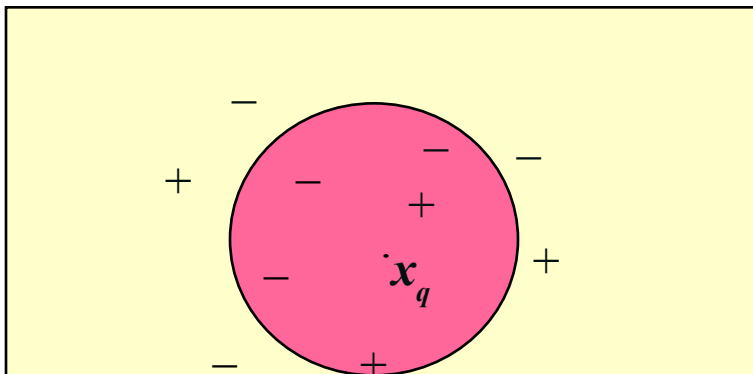
❖ Disadvantages

- Assumption: **class conditional independence**, therefore loss of accuracy
- Practically, **dependencies exist among variables**, e.g.:
 - Hospitals Patients Profile: age, family history, etc.
 - Symptoms: fever, cough etc.,
 - Disease: lung cancer, diabetes, etc.
- Dependencies among these **cannot be modeled by Naïve Bayes Classifier**

❖ How to deal with these dependencies? **Bayesian Belief Networks**

The k -Nearest Neighbor (k -NN) Algorithm

- ❖ All instances correspond to points in the n -D space
- ❖ The nearest neighbor are defined in terms of Euclidean distance, $\text{dist}(\mathbf{X}_1, \mathbf{X}_2)$
- ❖ Target function could be discrete- or real- valued
- ❖ For discrete-valued, k -NN returns the most common value (= majority vote) among the k training examples nearest to x_q
- ❖ Voronoi diagram: the decision surface induced by 1-NN for a typical set of training examples



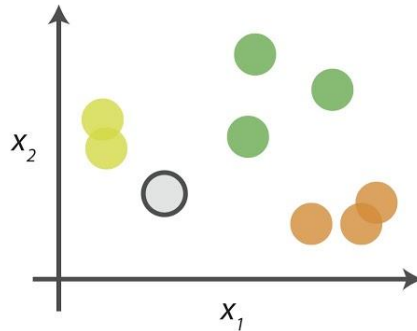
Discussion on the k -NN Algorithm

k -NN for real-valued prediction for a given unknown tuple

- Returns the mean values of the k nearest neighbors
- ❖ Distance-weighted nearest neighbor algorithm
 - Weight the contribution of each of the k neighbors according to their distance to the query x_q
 - Give greater weight to closer neighbors $w \equiv \frac{1}{d(x_q, x_i)^2}$
- ❖ Robust to noisy data by averaging k -nearest neighbors
- ❖ Curse of dimensionality: distance between neighbors could be dominated by irrelevant attributes
 - To overcome it, axes stretch or elimination of the least relevant attributes
- ❖ Categorical (non-numerical) attributes, e.g., color, are difficult to handle

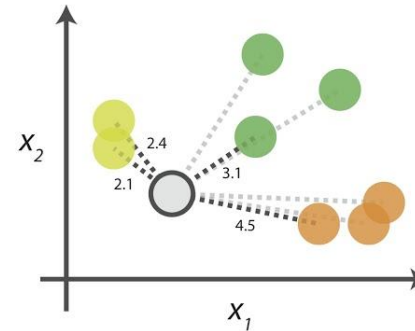
k-NN Algorithm

0. Look at the data



Say you want to classify the grey point into a class. Here, there are three potential classes - lime green, green and orange.

1. Calculate distances



Start by calculating the distances between the grey point and all other points.

2. Find neighbours

Point Distance		
 	2.1	→ 1st NN
 	2.4	→ 2nd NN
 	3.1	→ 3rd NN
 	4.5	→ 4th NN

Next, find the nearest neighbours by ranking points by increasing distance. The nearest neighbours (NNs) of the grey point are the ones closest in dataspace.

3. Vote on labels

Class	# of votes	
	2	→ Class  wins the vote! Point  is therefore predicted to be of class  .
	1	
	1	

Vote on the predicted class labels based on the classes of the k nearest neighbours. Here, the labels were predicted based on the $k=3$ nearest neighbours.

What Is Prediction?

- ❖ (Numerical) prediction is similar to classification
 - construct a model
 - use model to predict continuous or ordered value for a given input
- ❖ Prediction is different from classification
 - Classification refers to predict categorical class label
 - Prediction models continuous-valued functions
- ❖ Major method for prediction: regression
 - model the relationship between one or more *independent* or **predictor** variables and a *dependent* or **response** variable
- ❖ Regression analysis
 - Linear and multiple regression
 - Non-linear regression
 - Other regression methods: generalized linear model, Poisson regression, log-linear models, regression trees

Linear Regression

- ❖ Linear regression: involves a response variable y and a single predictor variable x

$$y = w_0 + w_1 x$$

where w_0 (y-intercept) and w_1 (slope) are regression coefficients

- ❖ Method of least squares: estimates the best-fitting straight line

$$w_1 = \frac{\sum_{i=1}^{|D|} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|} (x_i - \bar{x})^2} \quad w_0 = \bar{y} - w_1 \bar{x}$$

- ❖ Multiple linear regression: involves more than one predictor variable
 - Training data is of the form $(\mathbf{X}_1, y_1), (\mathbf{X}_2, y_2), \dots, (\mathbf{X}_{|D|}, y_{|D|})$
 - Ex. For 2-D data, we may have: $y = w_0 + w_1 x_1 + w_2 x_2$
 - Solvable by extension of least square method or using SAS, S-Plus
 - Many nonlinear functions can be transformed into the above

Example: Linear Regression

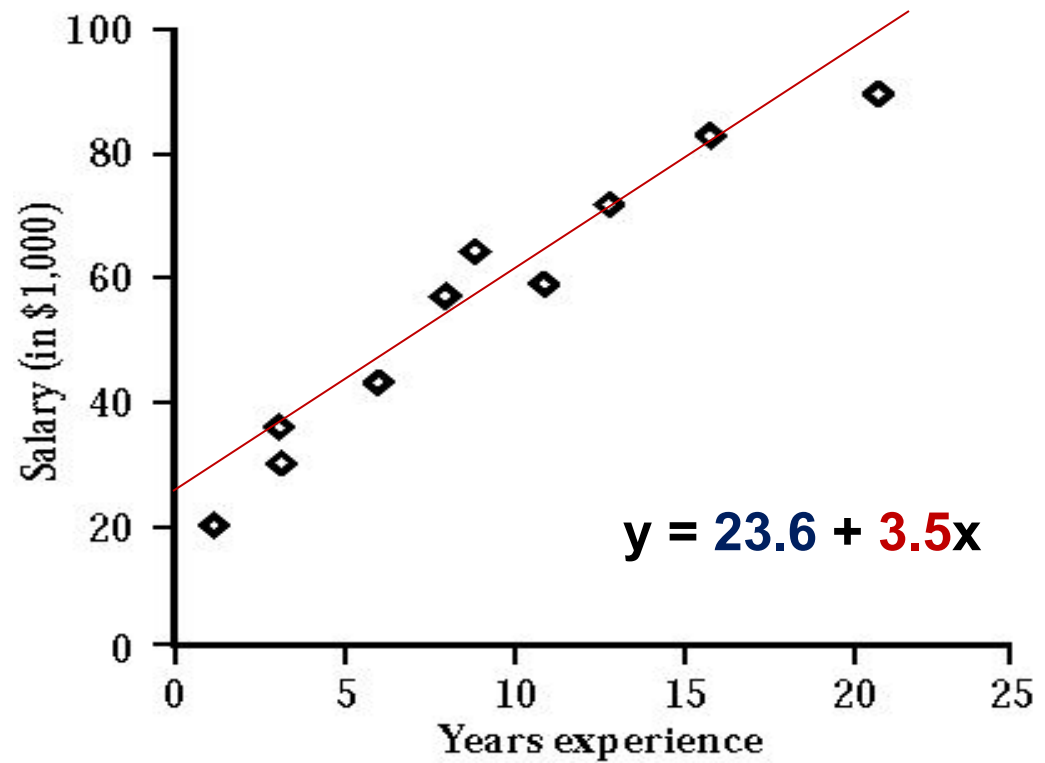
X (Years experience)	Y (Salary)
3	30K
8	57K
9	64K
13	75K
3	36K
6	43K
11	59K
21	90K
1	20K
16	83K

❖ Given data

$$\bar{x} = 9.1 \quad \bar{y} = 55.4$$

$$w_1 = \frac{(3-9.1)(30-55.4) + \dots + (16-9.1)(83-55.4)}{(3-9.1)^2 + \dots + (16-9.1)^2}$$

$$w_0 = 55.4 - (3.5)(9.1) = 23.6$$



Example: Linear Regression Analysis

X (Years experience)	Y (Salary)	Y (Predicted) [Y=23.6 +3.5X]	Mean Absolute Error	Root Mean Squared Error
3	30K	34.1K	abs(Y-Yp)=4.1	Sqr(Y-Yp)=16.81
8	57K	51.6K	5.4	29.16
9	64K	55.1K	8.9	79.21
13	75K	69.1K	5.9	34.81
3	36K	34.1K	1.9	3.61
6	43K	44.6K	1.6	2.56
11	59K	62.1K	3.1	9.61
21	90K	97.1K	7.1	50.41
1	20K	27.1K	7.1	50.41
16	83K	79.6K	3.4	11.56
			4.85	5.37

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (P_i - O_i)^2}{n}}$$



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Q & A

